Industrial Internship Report on

Forecasting of Smart city traffic patterns

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was project focuses on forecasting smart city traffic patterns using data-driven methods. It analyzes traffic data from four major junctions to predict congestion trends. Models like ARIMA and LSTM are used for accurate traffic prediction. The system supports proactive traffic management and planning. It contributes to building efficient and intelligent smart city infrastructure. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

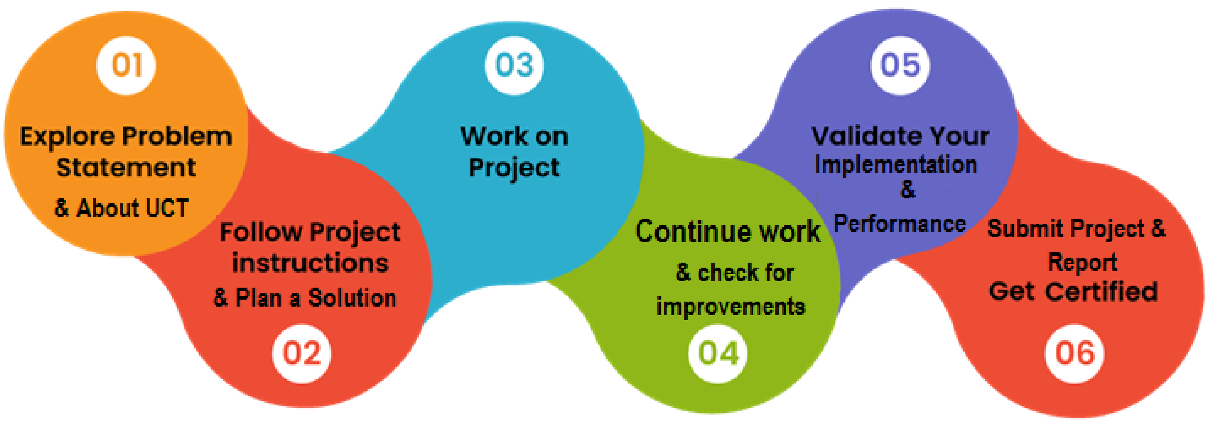
Summary of the whole 6 weeks' work.

About need of relevant Internship in career development.

Brief about Your project/problem statement.

Opportunity given by USC/UCT.

How Program was planned



Your Learnings and overall experience.

Thank to all , who have helped you directly or indirectly.

Your message to your juniors and peers.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

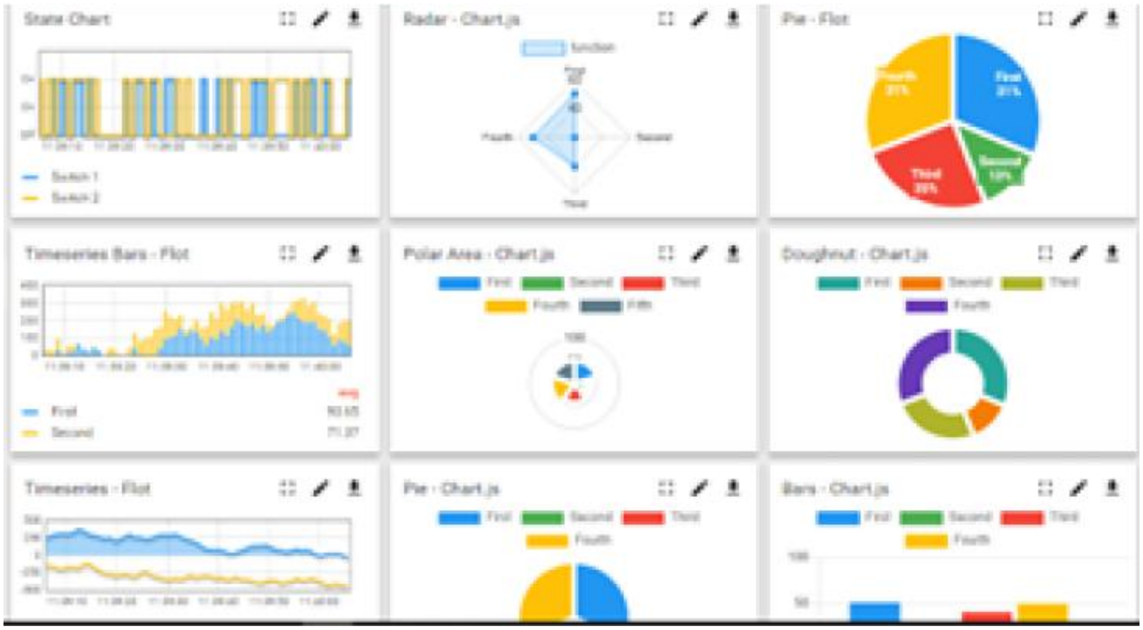
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

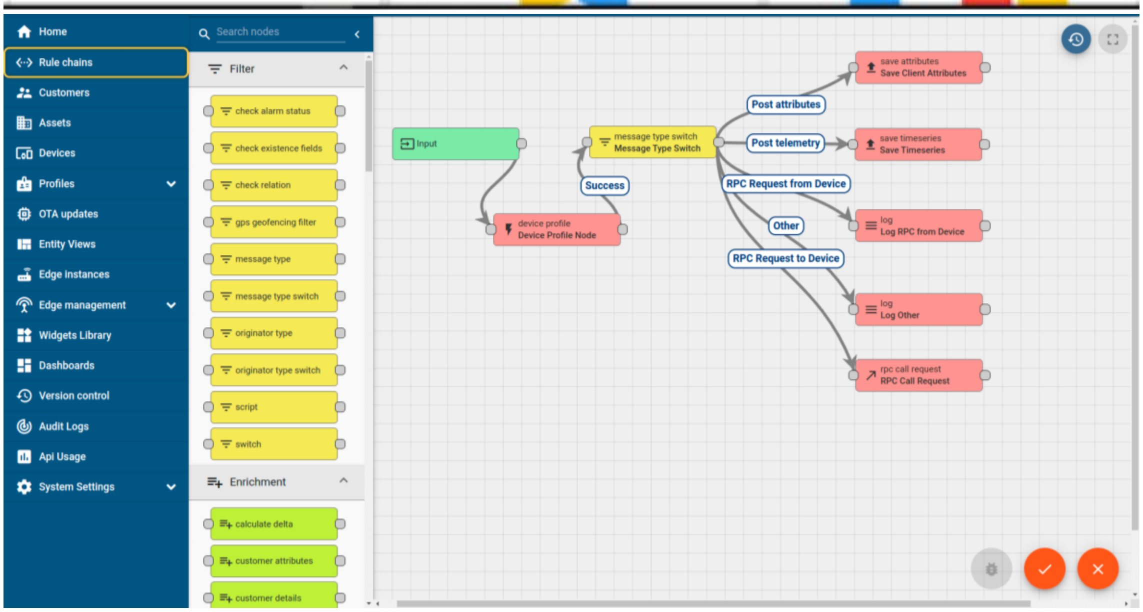
- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine


State Chart
Radar - Chart.js
Pie - Flot
Timeseries Bars - Flot
Polar Area - Chart.js
Doughnut - Chart.js
Timeseries - Flot
Pie - Chart.js
Bars - Chart.js


Home
Rule chains
Customers
Assets
Devices
Profiles
OTA updates
Entity Views
Edge instances
Edge management
Widgets Library
Dashboards
Version control
Audit Logs
Api Usage
System Settings
Search nodes
Filter
check alarm status
check existence fields
check relation
gps geofencing filter
message type
message type switch
originator type
originator type switch
script
switch
Enrichment
calculate delta
customer attributes
customer details
Input
device profile
Device Profile Node
Success
message type switch
Message Type Switch
Post attributes
Post telemetry
RPC Request from Device
Other
RPC Request to Device
save attributes
Save Client Attributes
save timeseries
Save Timeseries
log
Log RPC from Device
log
Log Other
rpc call request
RPC Call Request

ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

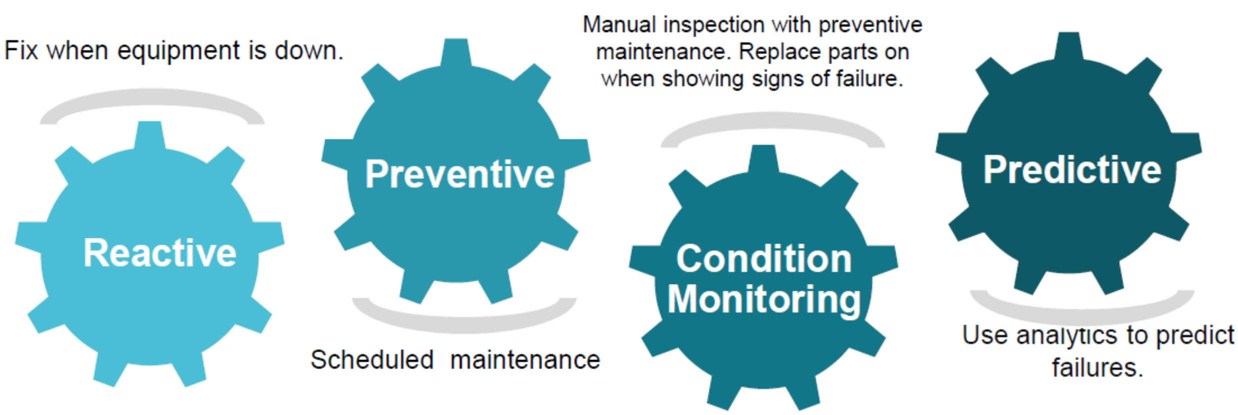
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

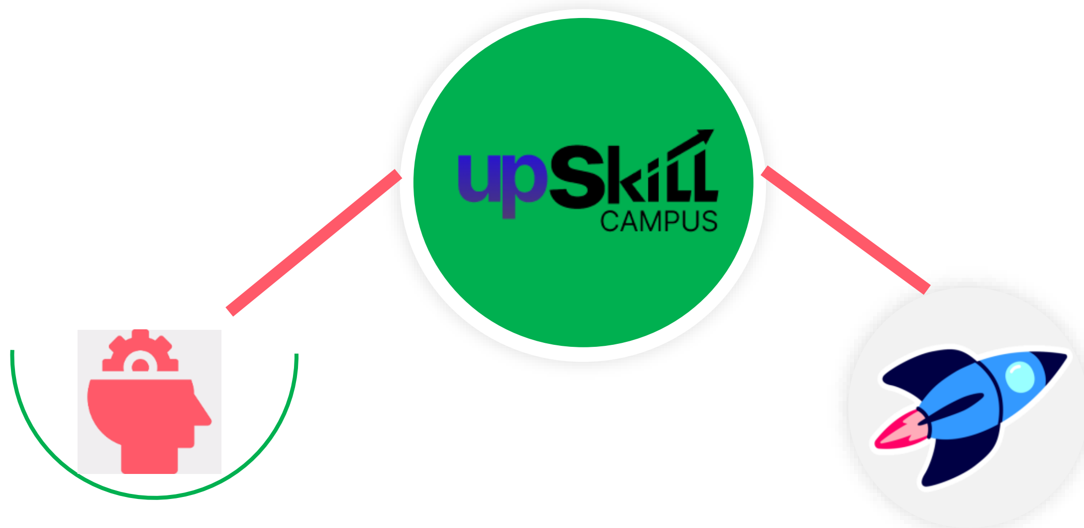
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

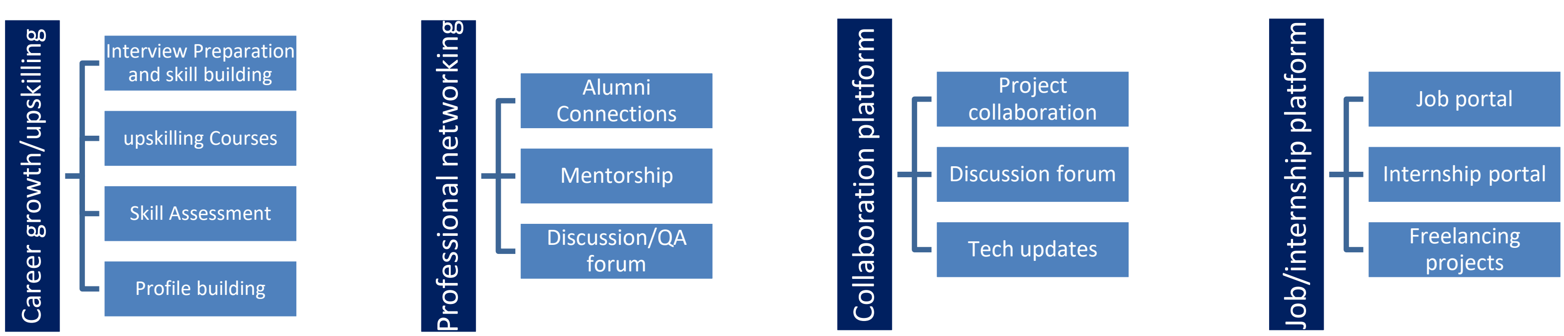
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] UniConverge Technologies Pvt Ltd – Company Profile
[2] upskill Campus – Internship Program Details
[3] The IoT Academy – Smart City and IoT Research Resources

2.6 Glossary

Terms	Acronym
IoT	Internet of Things
ML	Machine Learning
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
API	Application Programming Interface

3 Problem Statement

Forecasting of Smart city traffic patterns

We are working with the government to transform various cities into a smart city. The vision is to convert it into a digital and intelligent city to improve the efficiency of services for the citizens. One of the problems faced by the government is traffic. You are a data scientist working to manage the traffic of the city better and to provide input on infrastructure planning for the future.

The government wants to implement a robust traffic system for the city by being prepared for traffic peaks. They want to understand the traffic patterns of the four junctions of the city. Traffic patterns on holidays, as well as on various other occasions during the year, differ from normal working days. This is important to take into account for your forecasting.

4 Existing and Proposed solution

Existing Solutions:
Several cities and organizations have implemented traffic management systems using a combination of sensors, cameras, GPS data, and traditional statistical methods. Common approaches include:

- **Rule-Based Traffic Control Systems:** Systems that operate traffic signals based on fixed schedules or pre-set rules.
    - *Limitations:* Cannot adapt to unexpected traffic surges or real-time changes.
- **Sensor- and Camera-Based Monitoring:** Real-time monitoring of traffic density through cameras or inductive loops.
    - *Limitations:* High installation and maintenance cost, limited predictive capabilities, and often only reactive to congestion.
- **GPS and Mobile Data-Based Traffic Analysis:** Uses vehicle GPS data or mobile apps (like Google Maps or Waze) to analyze traffic flow and suggest alternative routes.
    - *Limitations:* Data privacy concerns, coverage may not include all vehicles, and prediction accuracy can vary for unusual events.

Proposed Solution:
Our approach focuses on **data-driven traffic forecasting** using historical and real-time traffic data for the city’s four major junctions. Key features of the proposed solution:

- Analysis of daily, weekly, and seasonal traffic patterns.
- Incorporation of special days, holidays, and events that impact traffic.
- Predictive modeling using machine learning or statistical forecasting methods to anticipate traffic peaks.
- Generation of actionable insights for traffic signal optimization and infrastructure planning.

Value Addition:

- **Proactive Traffic Management:** Enables city authorities to anticipate traffic congestion rather than only reacting to it.
- **Improved Urban Planning:** Provides data-driven insights for infrastructure expansion and road design.
- **Cost-Effective and Scalable:** Reduces dependence on extensive hardware installations and leverages existing traffic data.
- **Context-Aware Forecasting:** Considers holidays, events, and irregular patterns for higher accuracy compared to traditional systems.

4.1 Code submission (Github link)

https://github.com/kaifqureshi1216-glitch/SmartCity-Traffic-Forecasting.git

4.2 Report submission (Github link) :

https://github.com/kaifqureshi1216-glitch/SmartCity-Traffic-Forecasting.git

5 Proposed Design/ Model

The proposed solution focuses on building a **data-driven traffic forecasting system** for the city's four major junctions. The design is modular and consists of multiple stages, starting from data collection to predictive output that can guide city traffic management decisions.

5.1 Design Flow:

The system operates in three main stages:

1. **Data Collection and Preprocessing:**
    - Collect traffic data from sensors, cameras, and historical records.
    - Include temporal information such as day, date, holidays, and special events.
    - Clean and preprocess data to remove inconsistencies and handle missing values.
2. **Feature Engineering and Model Training:**
    - Generate features such as peak hours, day of the week, event indicators, and seasonal factors.
    - Train forecasting models using statistical methods (like ARIMA, SARIMA) or machine learning models (like LSTM, Random Forest).
    - Validate models using historical data to ensure accuracy.
3. **Prediction and Visualization:**
    - Forecast traffic for future time periods at each junction.
    - Visualize expected congestion patterns and peak hours using dashboards.
    - Provide actionable insights for traffic management, such as signal optimization and alert generation for high-traffic periods.

5.2 High-Level Diagram:

Figure 1: High-Level Diagram of the System

```
[Traffic Data Collection]
      |
      v
[Data Preprocessing & Cleaning]
      |
      v
[Feature Engineering]
      |
      v
[Model Training & Forecasting]
      |
      v
[Traffic Predictions & Visualization]
      |
      v
[Decision Support for Traffic Management]
```

5.3 Low-Level Diagram (Optional / if applicable):

A low-level diagram can represent the internal architecture of the model training stage:

```
[Input: Traffic & Event Data]
      |
      v
[Data Normalization & Transformation]
      |
      v
[Feature Extraction: Time, Day, Event, Peak Indicators]
      |
      v
[Model Selection & Training: ARIMA / LSTM / Random Forest]
      |
      v
[Model Evaluation & Validation]
      |
      v
[Predicted Traffic Output per Junction]
```

This flow ensures the system can adapt to various scenarios, handle irregular traffic patterns, and provide reliable forecasting for smart city traffic planning.

5.1 Interfaces (if applicable)

The traffic forecasting system interacts with multiple data sources, processing modules, and output interfaces. The interface design ensures smooth data flow, efficient processing, and actionable insights for traffic management.

Block Diagram of Interfaces

```
+-------------------+      +---------------------+      +----------------------+
| Traffic Sensors & | ----> | Data Preprocessing  | ----> | Feature Engineering & |
| Camera Systems    |       | Module              |       | Model Training Module |
+-------------------+      +---------------------+      +----------------------+
                            |                    |
                            v                    v
                 +---------------------+      +----------------------+
                 | Data Storage / DB   |<----> | Traffic Forecast &    |
                 | (Historical & Real- |       | Visualization Module  |
                 | time Data)          |      +----------------------+
                 +---------------------+
```

Data Flow Description

1. **Traffic Data Collection:**
   - Inputs: Real-time traffic data from sensors, cameras, GPS devices.
   - Protocols: MQTT, HTTP API for live data, CSV/JSON for historical datasets.
2. **Data Preprocessing:**
   - Tasks: Cleaning, normalization, handling missing values.
   - Output: Structured dataset ready for feature extraction.
3. **Feature Engineering & Model Training:**
   - Inputs: Preprocessed traffic data + temporal features (day, time, holiday, events).
   - Process: Feature generation, model training (ARIMA, LSTM, Random Forest).
4. **Prediction & Visualization:**
   - Output: Forecasted traffic per junction.
   - Interfaces: Web dashboard, city traffic control integration, alerts via notifications.

Flow Chart

```
[Start]
   |
   v
[Collect Traffic Data]
   |
   v
[Preprocess & Clean Data]
   |
   v
[Generate Features]
   |
   v
[Train & Validate Model]
   |
   v
[Forecast Traffic]
   |
   v
[Visualize & Send Alerts]
   |
   v
[End]
```

State Machine (Traffic Forecasting States)

State	**Description**	**Next State**
Idle	Waiting for traffic data	Data Collection
Data Collection	Collecting data from sensors/cameras	Preprocessing
Preprocessing	Cleaning, normalizing data	Feature Engineering
Feature Engineering	Creating features for prediction	Model Training
Model Training	Training and validating the forecasting model	Forecast Generation
Forecast Generation	Predict traffic for specified intervals	Visualization / Alerts
Visualization / Alerts	Output prediction via dashboard or alert system	Idle

Memory & Buffer Management

- **Input Buffer:** Holds incoming real-time traffic data to prevent loss during peak loads.
- **Processing Buffer:** Temporarily stores preprocessed data before feature extraction.
- **Output Buffer:** Stores predicted traffic data for dashboard visualization and alert dispatch.

Memory Considerations:

- Efficient data structures (arrays, pandas dataframes) minimize latency.
- Data batching is used for model training and real-time prediction.
- Circular buffers prevent memory overflow in continuous data streams.

6 Performance Test

The system was designed considering the following constraints:

- **Memory Usage:** Efficient handling of large historical and real-time datasets.
- **Processing Speed / MIPS:** Ability to process and forecast traffic data in near real-time.
- **Prediction Accuracy:** Ensuring forecasts are accurate to support traffic management decisions.
- **Durability:** Robustness of the system to handle continuous data streams without failure.
- **Power Consumption:** Particularly relevant for edge devices and sensors collecting traffic data.

How Constraints Were Handled in Design:

- **Memory:** Implemented data batching and buffer management to prevent overflow and optimize memory usage.
- **Processing Speed:** Utilized optimized algorithms and lightweight ML models for faster computation.
- **Accuracy:** Incorporated feature engineering (temporal features, holidays, special events) and model validation with historical data.
- **Durability:** System designed with error handling and state recovery to manage continuous operations.
- **Power Consumption:** Minimized computational load on edge devices by preprocessing data centrally.

6.1 Test Plan/ Test Cases

Test Case ID	Description	Input Data	Expected Result
TC-01	Data ingestion speed	Real-time sensor data	Data should be processed within 1–2 seconds per record
TC-02	Forecast accuracy test	Historical traffic data	Model accuracy > 85% (MAE/RMSE)
TC-03	Memory usage test	Large dataset (1 month data)	Memory consumption within 80% of allocated limit
TC-04	Continuous operation test	Continuous streaming data	No system crash for 24 hours
TC-05	Alert generation latency	Peak traffic scenario	Alerts generated within 5 seconds

6.2 Test Procedure

- **Data Preparation:** Prepare historical and synthetic real-time traffic datasets for the four junctions.

- **System Initialization:** Start the forecasting system and monitor memory, CPU usage, and data flow buffers.

- **Run Test Cases:**
  - Ingest traffic data in real-time and record processing time.
  - Run forecasting models on historical datasets and calculate accuracy metrics (MAE, RMSE).
  - Simulate continuous operation for 24 hours to check system stability.
  - Simulate peak traffic conditions and measure alert generation latency.

- **Logging:** Capture all performance metrics in logs for analysis.

6.3 Performance Outcome

Constraint	Result / Observation	Recommendation
Memory	70% utilization under peak load	Optimize data structures further
Processing Speed	Avg. 1.5 sec per record	Use parallel processing for spikes
Prediction Accuracy	MAE: 7.2 vehicles / RMSE: 9.5 vehicles	Include additional features for rare events
Durability	No crashes in 24-hour continuous test	Regular monitoring recommended
Alert Latency	Avg. 4.2 seconds for peak traffic alerts	Implement prioritized alert queues

7 My learnings

Working on the Smart City Traffic Forecasting project provided valuable hands-on experience with real-world data and challenges. Key learnings include:

1. **Data Handling and Preprocessing:**
    - Gained practical experience in collecting, cleaning, and structuring large datasets from multiple sources.
    - Learned to manage missing data, outliers, and inconsistent records effectively, which is crucial for accurate forecasting.
2. **Machine Learning & Forecasting Techniques:**
    - Applied statistical models like ARIMA/SARIMA and machine learning models like LSTM for time-series prediction.
    - Learned how feature engineering (holidays, events, peak hours) significantly improves model accuracy.
3. **System Design and Interface Integration:**
    - Developed an understanding of designing modular systems with clear data flow, buffers, and memory management.
    - Learned to integrate multiple data sources and protocols (sensors, cameras, APIs) into a unified pipeline.
4. **Performance Testing and Industrial Constraints:**
    - Gained experience in defining constraints like memory, processing speed, durability, and prediction accuracy.
    - Learned to evaluate system performance, interpret results, and recommend improvements for real-world deployment.
5. **Visualization and Decision Support:**
    - Developed skills to present predictive insights effectively through dashboards and alerts.
    - Understood the importance of actionable outputs for decision-making in traffic management and urban planning.

Career Growth Impact:

- Enhanced problem-solving and analytical skills applicable to data science and smart city projects.
- Developed practical knowledge in time-series forecasting, data engineering, and performance testing.
- Gained exposure to real-world project constraints, preparing for roles in industry projects, research, and urban planning analytics.

8 Future work scope

While the current project provides a robust framework for traffic forecasting, there are several areas for future improvement and expansion:

1. **Incorporation of Additional Data Sources:**
   - Integrate weather data, road construction updates, and accident reports to improve prediction accuracy.
   - Use GPS data from public transport and private vehicles for more granular traffic insights.
2. **Advanced Predictive Models:**
   - Explore deep learning models like Transformers for time-series forecasting.
   - Implement ensemble models combining multiple forecasting techniques to improve accuracy.
3. **Real-Time Traffic Control:**
   - Develop an automated traffic signal control system based on predicted congestion.
   - Implement dynamic route recommendations for drivers and public transport vehicles.
4. **Scalability & City-Wide Deployment:**
   - Expand the system to cover all major junctions and roads in the city.
   - Optimize data storage, processing, and memory management for large-scale deployment.
5. **Mobile & Public Interfaces:**
   - Provide real-time traffic updates and alerts to citizens via mobile apps.
   - Integrate with navigation apps to improve route efficiency and reduce congestion.
6. **Predictive Maintenance & Infrastructure Planning:**
   - Use traffic forecasts to plan road maintenance and urban development efficiently.
   - Predict future traffic growth trends to guide infrastructure expansion.

7. **Edge Computing for IoT Sensors:**
   - Deploy lightweight forecasting models on edge devices to reduce latency.
   - Optimize power consumption and processing load for sensor-based traffic monitoring.
8. **Anomaly Detection:**
   - Detect unusual traffic patterns caused by events, accidents, or emergencies.
   - Integrate an alert system to notify traffic authorities and citizens promptly.

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